EXPERIMENT 5
Report Generation Using Relational and Multidimensional Data Using Microsoft Power BI

1. AIM
To generate relational and multidimensional reports using Microsoft Power BI by utilizing transactional and dimensional data models, enabling effective business analysis, reporting, and decision-making.

2. OBJECTIVES
- To understand the concepts of relational and multidimensional reporting in Business Intelligence.
- To generate relational reports using transactional business data.
- To create multidimensional reports using hierarchies and OLAP concepts.
- To apply sorting, filtering, grouping, and aggregation to business data.
- To perform drill-down and drill-up operations for detailed analysis.
- To create comparative visualizations for analyzing sales, profit, and customer performance.
- To design interactive dashboards using KPI cards and slicers.
- To interpret business insights from reports and visualizations.
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3. TOOLS / SOFTWARE REQUIRED
- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

Dataset Source: Global Superstore Dataset (Kaggle)
https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

4. THEORY
Business Intelligence (BI) reporting is the process of transforming raw business data into meaningful information for analysis and decision-making. Microsoft Power BI is a Business Intelligence tool that enables users to connect, transform, analyze, and visualize data through interactive reports and dashboards.

Relational Reporting: Relational reports display detailed transactional records in a tabular format. They are useful for examining individual orders, customers, products, sales, and profits. Power BI provides table visuals, sorting, and filtering features to generate relational reports.

Multidimensional Reporting: Multidimensional reports allow data to be analyzed across multiple dimensions, such as Region, State, City, Product, Customer, and Time. Matrix visuals and hierarchies help users summarize and explore business performance at different levels.

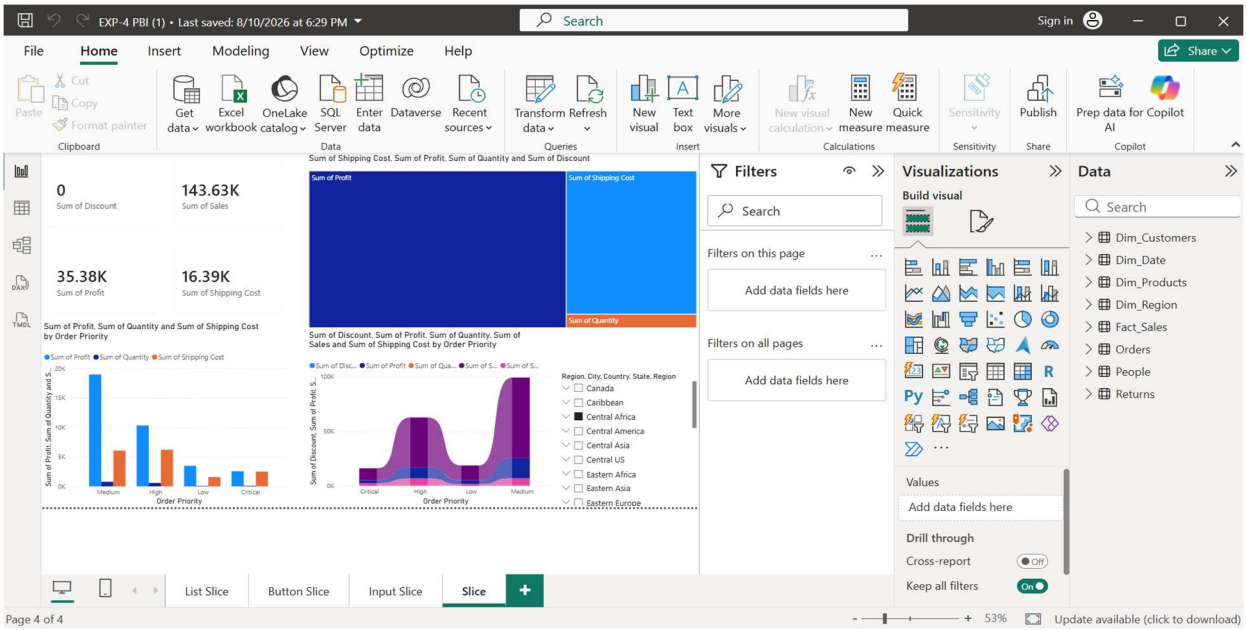
OLAP (Online Analytical Processing): OLAP enables users to analyze data from different perspectives. Drill-down provides more detailed information, while drill-up returns to a higher level of summarized information.

Data Visualization: Power BI supports bar charts, column charts, line charts, tables, matrices, and KPI cards to represent business data effectively.

Interactive Dashboard: An interactive dashboard combines multiple visualizations and slicers into a single report page. Slicers allow users to filter information by dimensions such as Region, Category, and Year.

5. PROCEDURE

Part A: Generate Relational Report



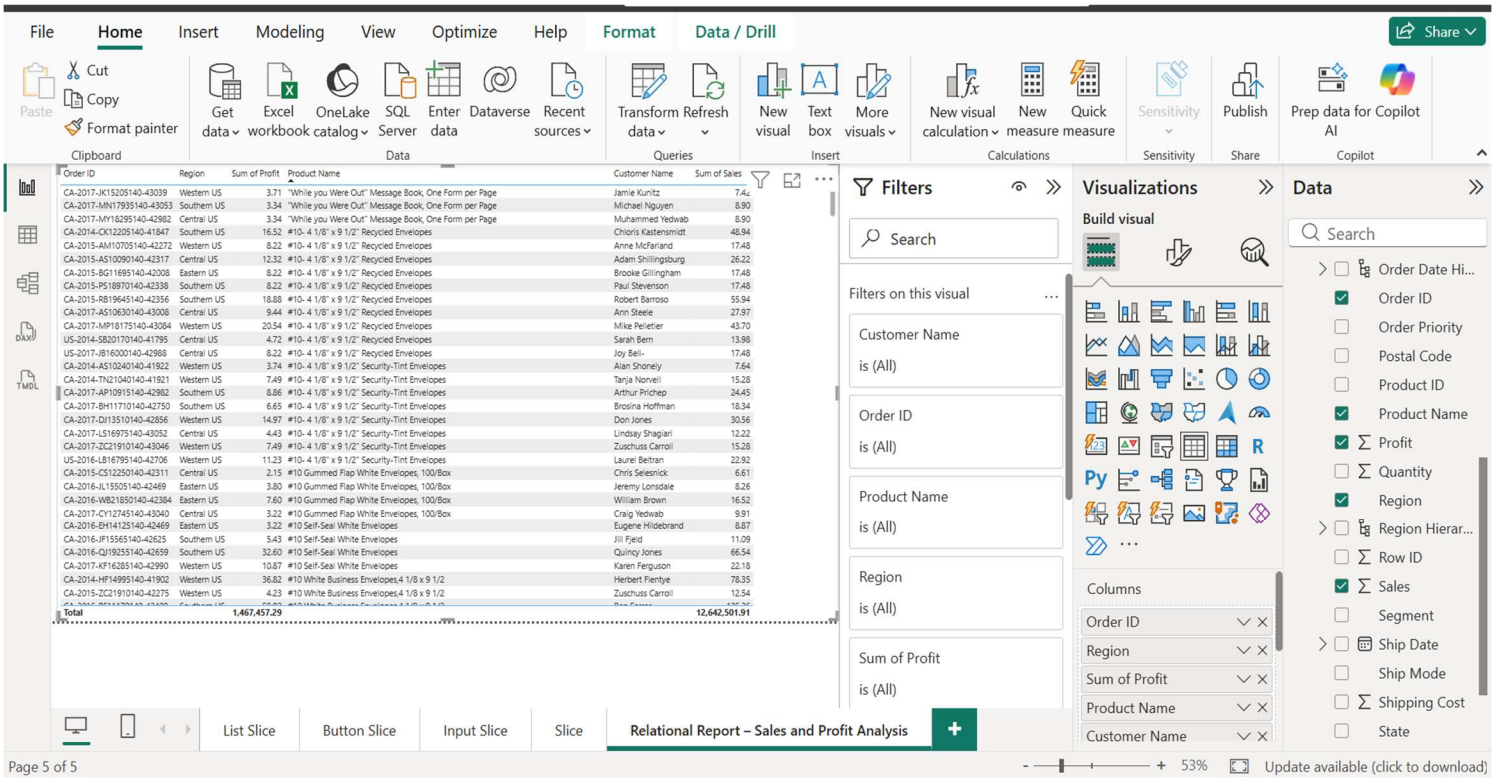
Step 1: Open the Power BI Project

Open Microsoft Power BI Desktop and load the previously created project containing the Global Superstore dataset.

Step 2: Verify the Dataset

Verify that the required tables, such as Orders, Fact_Sales, Dim_Customers, and Dim_Region, are available in the Data pane.

Step 3: Create a Table Visual



Select the Table visual from the Visualizations pane and place it on the report canvas.

Step 4: Add Required Fields

Add the following fields to the table visual:

- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit
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Step 5: Apply Sorting and Filtering

Sort the Sales column in descending order to display higher sales values first. Apply a Region filter to analyze records for a selected geographical region.

Step 6: Export the Report

Select the table visual, click the three-dot menu, and choose Export data. Export the data to a supported format, such as CSV, and open it in Microsoft Excel if required.

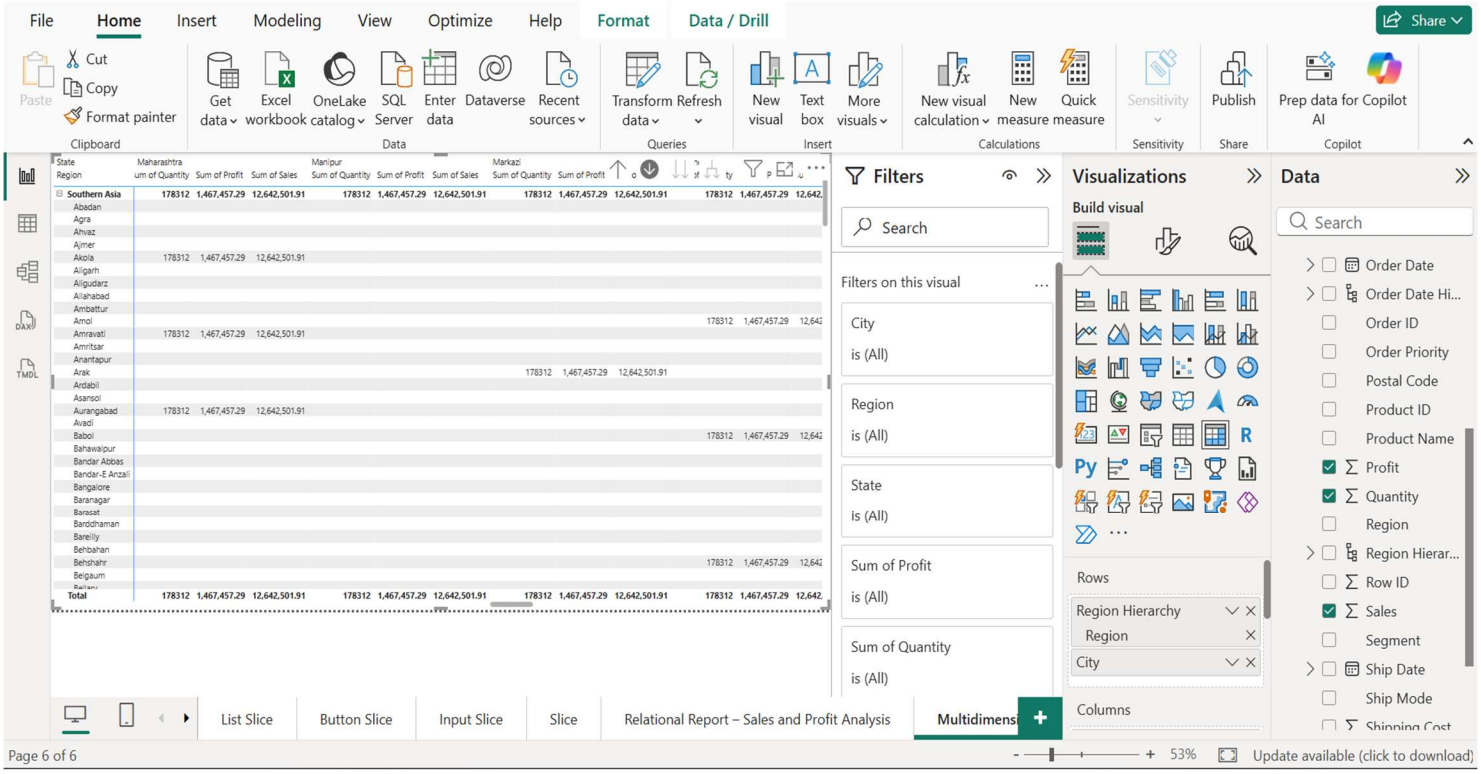
Step 7: Record Observations

Observe the transactional details and identify differences in sales and profit across orders, customers, and regions.

Part B: Generate Multidimensional Report

Step 8: Create a Hierarchy

In the Dim_Region table, create a hierarchy named Region Hierarchy using the following order:



Region → State → City

Step 9: Insert a Matrix Visual

Select the Matrix visual from the Visualizations pane and place it on the report canvas.

Step 10: Add the Hierarchy and Measures

Add Region Hierarchy to the Rows section of the Matrix visual.
Add the following measures or numeric fields to the Values section:

- Sales
- Profit
- Quantity

Ensure that Sales, Profit, and Quantity are aggregated appropriately, usually using Sum.

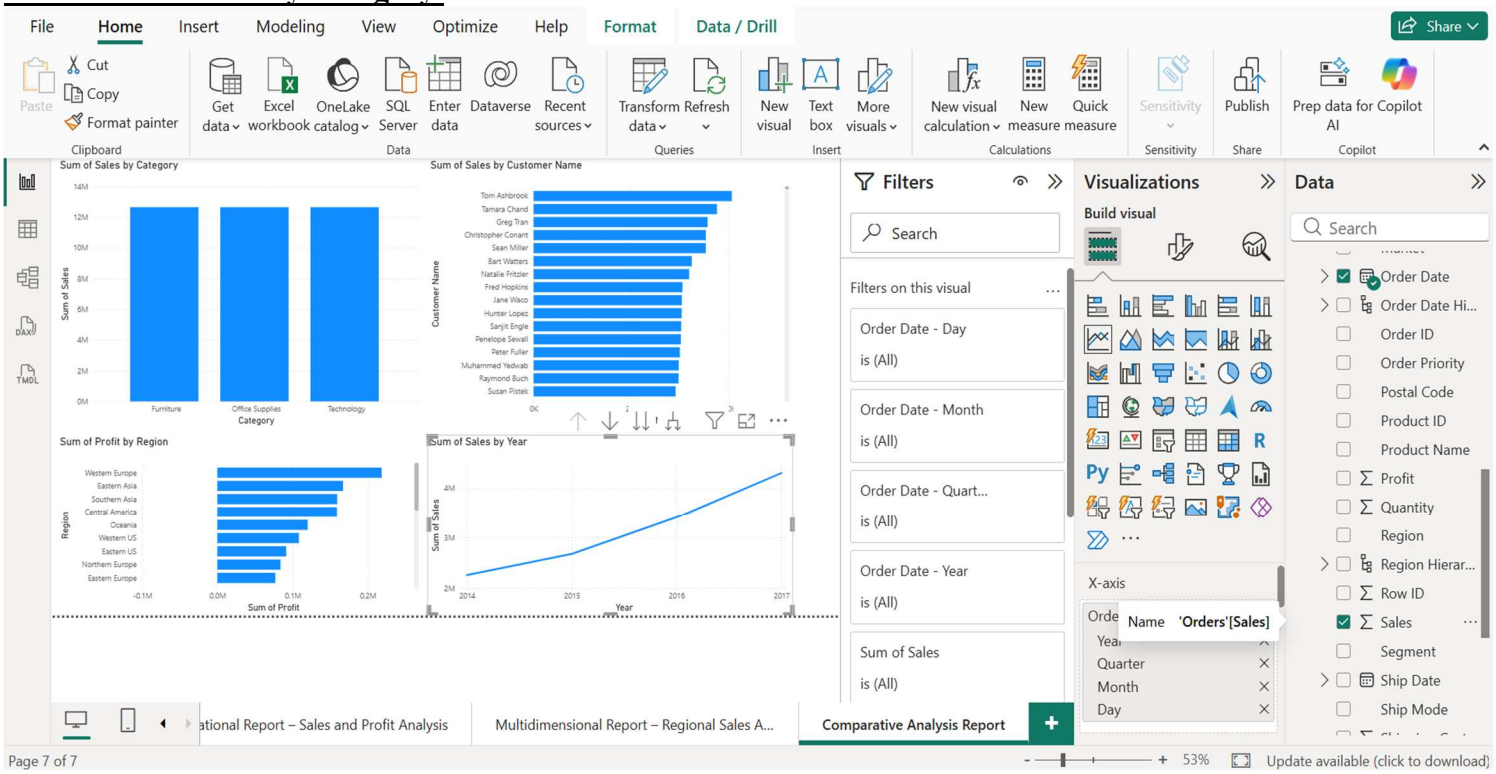
Step 11: Perform Drill-down and Drill-up
Use the hierarchy controls to navigate from Region to State and then City. Use Drill-up to return to higher levels of aggregation.

Step 12: Analyze Aggregated Performance
Compare sales, profit, and quantity across different geographical levels to identify regional differences in business performance.

Part C: Generate Comparative Analysis Report

Step 13: Create Sales by Category Chart
Insert a Clustered Column Chart. Add Category to the X-axis and Sales to the Y-axis. Set the aggregation of Sales to Sum.

Set the title to **Sales by Category**.



Step 14: Create Profit by Region Chart
Insert a Clustered Bar Chart. Add Region to the Y-axis and Profit to the X-axis. Set Profit aggregation to Sum.
Set the title to **Profit by Region**.

Step 15: Create Top 10 Customers by Sales Chart
Insert a Clustered Bar Chart. Add Customer Name to the Y-axis and Sales to the X-axis.
Apply a Top N filter with the following settings:

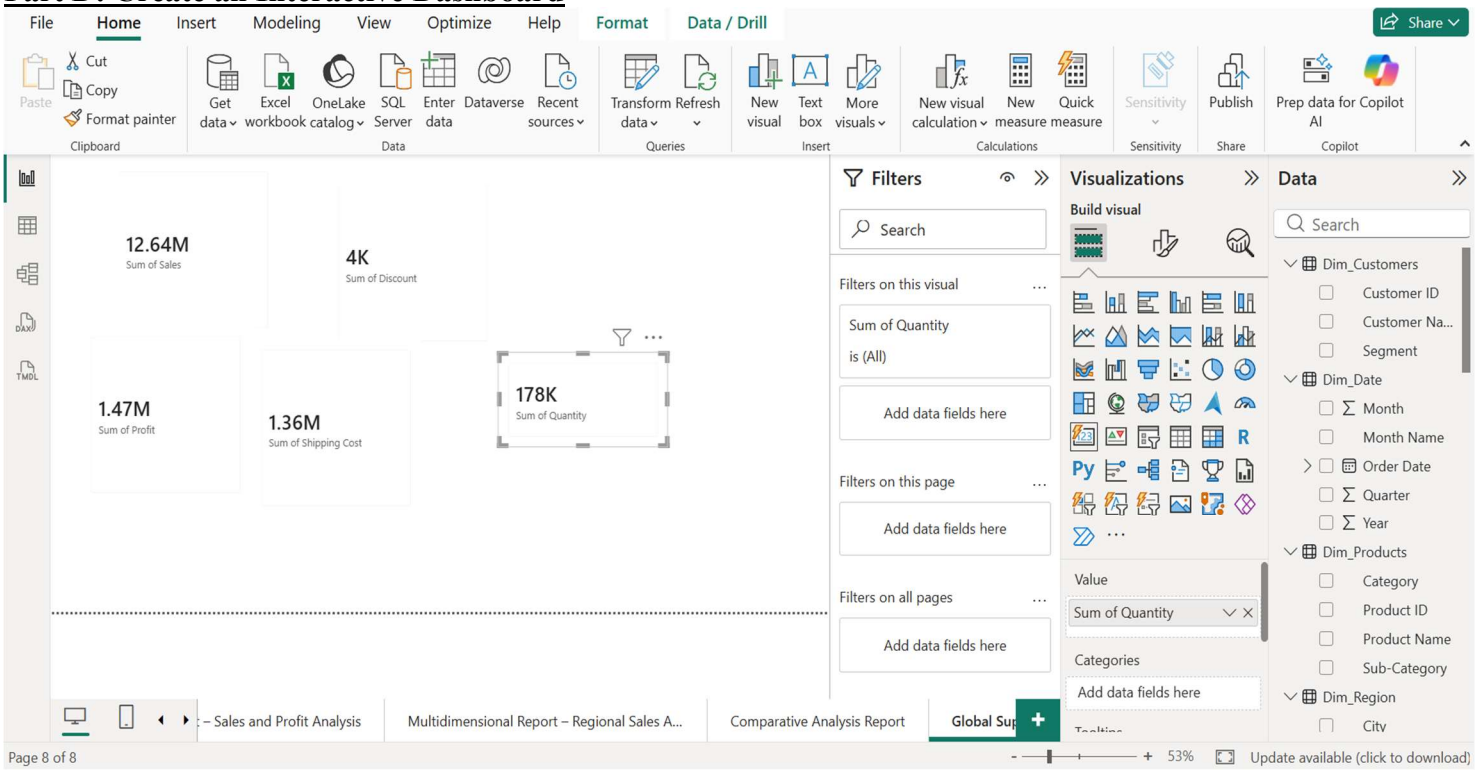
- Filter type: Top N
- Show items: Top 10
- By value: Sales
- Aggregation: Sum of Sales

Apply the filter and sort the chart by Sales in descending order.
Set the title to **Top 10 Customers by Sales**.

Step 16: Create Year-wise Sales Trend
Insert a Line Chart. Add Order Date to the X-axis and Sales to the Y-axis.
Select the Year level of the Date hierarchy and aggregate Sales using Sum.
Set the title to **Year-wise Sales Trend**.

Step 17: Compare the Visualizations
Compare the results of the four visualizations to understand sales distribution by category, profitability by region, customer contribution, and yearly sales trends.

Part D: Create an Interactive Dashboard



Step 18: Create Total Sales KPI Card
Insert a Card visual and add Sales with Sum aggregation.
Set the title to **Total Sales**.

Step 19: Create Total Profit KPI Card
Insert another Card visual and add Profit with Sum aggregation.
Set the title to **Total Profit**.

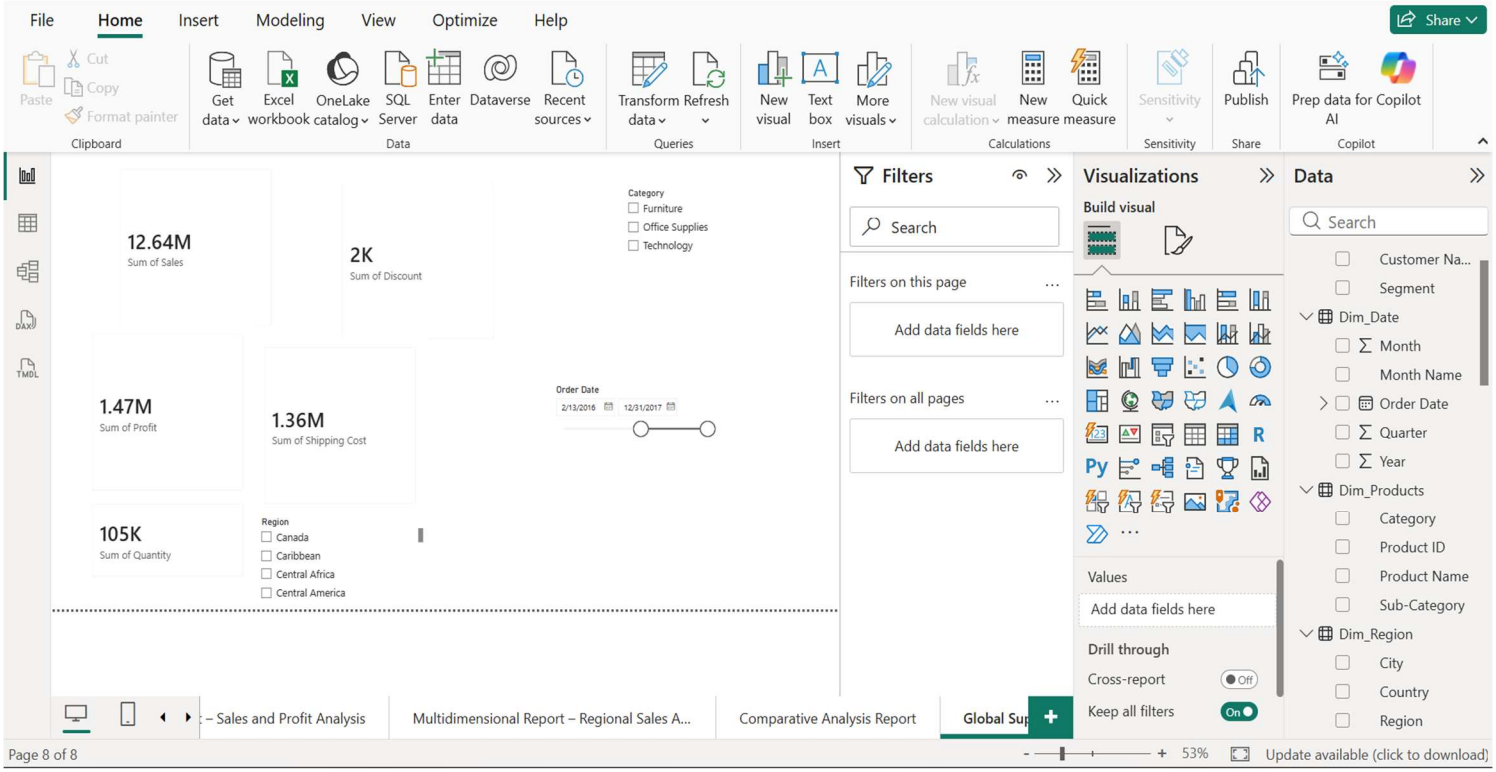
Step 20: Create Total Orders KPI Card
Insert another Card visual and add Order ID.
Set the aggregation to Distinct Count to count unique orders rather than individual product rows.
Set the title to **Total Orders**.

Step 21: Add Region Slicer
Insert a Slicer visual and add Region from the appropriate region dimension table.
Set the title to **Filter by Region**.

Step 22: Add Category Slicer

Insert another Slicer visual and add Category from the appropriate product or sales table.

Set the title to **Filter by Category**.



Step 23: Add Year Slicer

Insert another Slicer visual and add Order Date. If using the Date hierarchy, select the Year level.

Set the title to **Filter by Year**.

Step 24: Arrange the Dashboard

Arrange the three KPI cards, slicers, and four charts into a clear and organized dashboard layout.

Set the dashboard title to:

Global Superstore – Business Performance Dashboard

Step 25: Verify Interactive Functionality

Select different values in the Region, Category, and Year slicers. Verify that the relevant charts and KPI cards update according to the selected filters.

If any visual does not respond as expected, check the visual interactions and data relationships.

Step 26: Save the Report

Save the completed Power BI project using Ctrl + S. Keep the report file in PBIX format for future editing and presentation.

6. OBSERVATIONS

1. A relational report was generated using a table visual to display transactional details such as Order ID, Customer Name, Product Name, Region, Sales, and Profit.
2. Sorting and filtering helped organize records and focus on selected regions.
3. The report data was successfully exported for further analysis.
4. A multidimensional report was generated using a Matrix visual and a Region → State → City hierarchy.
5. Drill-down and drill-up operations enabled analysis at different geographical levels.
6. The Sales by Category chart helped compare sales across product categories.
7. The Profit by Region chart enabled comparison of regional profitability.
8. The Top 10 Customers by Sales chart highlighted customers contributing the highest sales.

9. The Year-wise Sales Trend chart represented changes in sales over time.
10. KPI cards displayed Total Sales, Total Profit, and Total Orders.
11. Region, Category, and Year slicers provided interactive filtering functionality.
12. The dashboard enabled consolidated analysis of business performance.

7. RESULT

Relational and multidimensional reports were successfully generated using Microsoft Power BI and the Global Superstore dataset. Sorting, filtering, hierarchy creation, drill-down, drill-up, comparative visualizations, KPI cards, and interactive slicers were implemented and tested. The final dashboard provided an integrated view of sales, profit, orders, customer performance, and regional trends, supporting effective business analysis and decision-making.

8. LEARNING OUTCOMES

- Learned to generate relational reports using transactional business data.
- Understood the concepts of multidimensional reporting and OLAP.
- Gained practical experience in creating hierarchies and Matrix visuals.
- Learned to apply sorting, filtering, grouping, and aggregation in Power BI.
- Gained experience in performing drill-down and drill-up operations.
- Learned to create comparative visualizations using bar, column, and line charts.
- Understood how KPI cards represent key business performance indicators.
- Learned to create interactive dashboards using slicers.
- Gained experience in interpreting business information from multiple visualizations.
- Understood how Business Intelligence reporting supports data-driven decision-making.